FISHES of SAHUL

VOLUME THIRTY SEVEN
NUMBER ONE

MARCH 2023

JOURNAL OF THE AUSTRALIA NEW GUINEA FISHES ASSOCIATION

Incorporated Registration No. A0027788J. ISSN: 0813-3778 (print) ISSN: 2205-9342 (online)

Melanotaenia ajamaruensis. Photo: Hans-Georg Evers

First record of the Critically Endangered Malanda Rainbowfish in the Russell-Mulgrave drainage basin, Atherton Tablelands.

Keith C. Martin & Susan Barclay
PO Box 520, Clifton Beach Qld 4879, keith.c.martin@outlook.com

Introduction
The Malanda Rainbowfish (*Melanotaenia* sp. nov "Malanda") is a currently undescribed rainbowfish endemic to the Wet Tropics region of north Queensland. The species has a restricted distribution in small, high elevation streams on the Atherton Tablelands and there is a significant threat to these populations from invasion of its habitat by the larger Eastern Rainbowfish (*Melanotaenia splendida splendida*) with which it can hybridise and thus damage the genetic integrity of the species as well as outright population swamping and replacement. Due to these and other factors, the species has been listed as "Critically Endangered" in both the IUCN Red List, and the Australian Government EPBC Act (Brown et. al 2019; Department of the Environment 2022).

The form has been known to aquarists for some time but has variously been confused with other related species such as Lake Eacham Rainbowfish (*Melanotaenia eachamensis*) and Utchee Rainbowfish (*Melanotaenia utcheensis*). It is only recently that this taxon has been recognised as distinct through genetic and meristic analysis, although it is yet to be formally described (Unmack et. al. 2016).

The species has to date been recorded only in some small upper tributaries of the North Johnstone River near the town of Malanda. Historically, it is likely that the species was widely distributed in this area, primarily in the Williams Creek and Ithaca River catchments to the west, plus a small unnamed creek on the eastern side of the North Johnstone River (Unmack et. al. 2016). It has since contracted, and continues to contract, to the upper reaches of these streams (Unmack et. al. 2016) and despite intensive searches of the region, only one other small population has been discovered since 2016, also in a Williams Creek tributary (M. Hammer, pers. comm).

All known populations occur in small upland tributary streams at elevations of 650–800 m. These streams are basalt-based, with basalt bedrock, boulders and cobbles with red alluvial substrates, and are often punctuated with small vertical drop waterfalls. All currently known sites are on private properties nearly all of which have been extensively cleared and modified for farming, although it is suspected that the original streamside habitat for this species would have been rainforest (Unmack et. al. 2016).

In this article we present the discovery of Malanda Rainbowfish at a rainforest location within a stream and drainage basin (Mulgrave-Russell) where it has not previously been recorded.

The Mulgrave-Russell Basin
Australia is divided into a series of freshwater drainage divisions, or provinces and within each are a series of more localised river


Male Malanda Rainbowfish from Butchers Creek. Photo: Keith Martin

Approaching Butchers Creek gorge. Photo: Susan Barclay

catchments, or "basins" (BOM 2001). In Queensland, the Wet Tropics Freshwater Biogeographic Province (WTFBP) consists of twelve eastern-flowing drainage basins from the Jeannie Basin in the north to the Black Basin, near Townsville in the south (Department of Environment and Science 2013). On the Atherton Tablelands, the headwaters of the Barron, Johnstone, Herbert and Tully basins are the primary freshwater resources of the region, and these are where the majority of freshwater fish species, including all the endemic species occur.

The eastern edge of the Atherton Tablelands is dominated by a range of forested mountains, including Queensland's highest, Mt. Bartle Frere, one of the wettest areas in Australia. Unlike almost all other drainages on the Atherton Tablelands, streams flowing from these mountains drain into the Mulgrave-Russell Basin, often forming cool, fast-flowing and very clear, sand/granite-based tributaries along the coastal plain south of Cairns including major streams such as Harvey Creek, Behana Creek and Babinda Creek. Rainbowfish species occurring in the lowland reaches and associated swamps include Eastern Rainbowfish, McCulloch's Rainbowfish (*Melanotaenia maccullochi*) and Cairns Rainbowfish (*Cairnsichthys rhombosomoides*).

Some of the upland tributaries of the Mulgrave-Russell Basin are sourced on the eastern edge of the Atherton Tablelands, at elevations of 700–800 m. They flow eastwards through dense tropical rainforest, often descending over high waterfalls (e.g. Windin Falls, Long Gadgarra Falls, West Mulgrave Falls and Caribou Falls) before reaching the Mulgrave River valley below. While the upper reaches of these streams have been partially cleared for agriculture in some areas, most of the catchments in this area are rainforest-clad and extremely difficult to access.

During the course of fish surveys in this area over the last ten years, the authors and co-researchers Michael Hammer and Peter Unmack have sampled some sites in these upper Mulgrave-Russell streams but in every case, we have either failed to find any rainbowfishes at all or have found probably translocated populations of Eastern Rainbowfish.

Malanda Rainbowfish in the Mulgrave-Russell Basin

In December 2021, the authors were granted permission to access a point along an upper Mulgrave River tributary stream, Butchers Creek, on private property adjacent to Gadgarra National Park. This area was in undisturbed rainforest.

Upon hiking into the site, we observed that Butchers Creek was a significant watercourse at that point, flowing strongly over basalt bedrock with riffle, small and deep pool habitat. Water quality data collected at the site were as follows: pH 8.0; E.C. 53 µs; TDS 34 ppm; and water temperature 21.6^{0}C. The average wetted width of the stream was 8 m, with depths of up to 2 m. Water flow was slow and there was about 40% rainforest cover.

Butchers Creek at collecting site. Photo: Keith Martin

Male (front) and female Malanda Rainbowfish from Butchers Creek Photo: Keith Martin

As is typical of basalt-based streams in this area, water turbidity was moderately high, with visibility for snorkel searching less than 1 metre. Nevertheless, snorkelling in a deep pool quickly revealed the presence of rainbowfish, so we set several baited box traps in shallow spots, and seine netted part of a deeper pool.

After several seine runs, we managed to catch our target number of 20 rainbowfish. Also present were numerous introduced Guppies (*Poecilia reticulata*) and in the traps, some Purple-spotted Gudgeon (*Mogurnda adspersa*). We needed about 20 rainbowfish to cover any losses during transit, so that we would have a sufficient male/female mix and good variation, both for genetic and meristic analysis and for live photography. We were also constrained by having to hike back to the vehicle some distance uphill, with the fish being transported in our backpacks in sealed plastic containers.

Straight out of the creek, these rainbowfish were obviously a small form. Some males exhibited a high dorsal fin and reddish coloured fins on a gold or brown/red body. First impressions were that these were an important find, as they were definitely not pure Eastern Rainbowfish, a commonly translocated native fish on the Atherton Tablelands. Based on our familiarity with all the Tablelands rainbowfish forms, our initial thoughts were that they were either Malanda or Lake Eacham Rainbowfish, leaning towards Malanda. There seemed to be some variation between individuals, indicating potential introgressions. None of these observations could be confirmed without genetic investigations, so it was important to get the specimens home, settled in, photographed, bred, and the wild fish sent off for scientific study.

Once settled into our aquariums, we set about getting quality photographs of the fish. These we sent to the experts (Peter Unmack & Michael Hammer) for their opinion. Like us, the consensus was no consensus – maybe Malanda, maybe Lake Eacham, but in any case an important find. We have a long-standing agreement with local aquarium wholesaler Cairns Marine to help our research project in transporting live fish. The fish were duly sent on to Michael Hammer at the NT Museum, where they were processed, preserved, and samples sent to Peter Unmack for genetic analysis. This process took some time, and it was not until October 2022 that we had some early results.

Based on the genetic analysis (using SNPs from genome-wide next generation sequencing), these fish turned out to be Malanda Rainbowfish. There could be some historic introgression with other species, possibly Lake Eacham or Eastern Rainbowfish, but these are early results so the full story is yet to be told. Purple-spotted Gudgeons collected at the same site turned out to be from the "Dirran" lineage (Briggs 2021) which were otherwise only known from the North Johnstone River (P. Unmack, pers. comm.).

Discussion

The discovery of Malanda Rainbowfish at a locality along Butchers Creek is quite significant given the conservation status of the species. The Butchers Creek site is approximately 10 km from the nearest known population of Malanda Rainbowfish so the range extension is relatively small. However, all previous records for this species come from North Johnstone River tributaries, in the Johnstone Basin, whereas Butchers Creek is part of the Mulgrave-

Butchers Creek showing shallow riffles, basalt rock base and undisturbed rainforest surrounds. Photo: Susan Barclay

Deep pool in Butchers Creek where we found Malanda Rainbowfish. Photo: Keith Martin

Keith climbing out of the Butchers Creek gorge with a backpack full of Malanda Rainbowfish. Photo: Susan Barclay

Male (front) and female Malanda Rainbowfish from Butchers Creek. Photo: Keith Martin

Russell Basin, therefore not only placing the species in a different stream system, but also in an entirely different drainage basin.

Butchers Creek has a fairly extensive catchment on the Atherton Tablelands, and while the upper reaches have extensive clearing, a significant extent of the creek flows through undisturbed rainforest, including much within a National Park. If, as is likely (but not yet confirmed), the species occurs well downstream of the current location it would be the only area where this species is protected within a National Park, as all localities in the North Johnstone catchment are on private property. It is also the only area where the species is present in continuous undisturbed aquatic and adjacent terrestrial environments. Butchers Creek traverses a distance of over seven kilometres of rainforest within Gadgarra

NT Museum voucher specimen, male Malanda Rainbowfish from Butchers Creek. Photo: Michael Hammer

Purple-spotted Gudgeon collected at Butchers Creek. Photo: Keith Martin

National Park before reaching a major waterfall (Long Gadgarra Falls) where it drops into the lowland Mulgrave River valley.

There is a possibility that the species may occur in other adjacent creeks of the area such as Toohey Creek to the north and, although there is some evidence that Eastern Rainbowfish may occur in Caribou Creek to the south (M. Hammer pers. comm.), these creeks have yet to be more thoroughly surveyed due to access and permitting constraints. Neither of these creeks are directly connected to Butchers Creek before reaching significant waterfall barriers.

The finding of Malanda Rainbowfish in a creek of the Mulgrave-Russell Basin raises the question of whether the species is native to that system or is a local translocation from a North Johnstone locality. The presence of a second species, Southern Purple Spotted Gudgeon with clear relationships to the North Johnstone River lends support to this being a natural distribution pattern. However, the presence of Guppies, and possible indication of introgressions from other species at the collecting site indicates that some introductions of fish may have occurred in the upper reaches of Butchers Creek at some stage. Based on indicators at the North Johnstone River sites where Malanda Rainbowfish occur, the habitat at Butchers Creek would appear to be ideal for this species.

Whether the species is native or translocated to this site is yet to be resolved but in a sense it really doesn't matter. In either case, it is a Critically Endangered species present in a quite remote locality, in original natural habitat and likely distributed over a large area of undisturbed stream protected within and adjacent to a National Park. These conditions bode well for the continued survival of this species.

Acknowledgements
Michael Hammer facilitated transfer of specimens and lodgement with the NT Museum. Peter Unmack conducted genetic analysis and provided early analysis of the results. This research was conducted under Qld Fisheries permit No: 212524 and University of Canberra Animal Ethics ID 6879.

References

Bureau of Meteorology 2001. Australian Drainage Divisions and River Basins. http://www.bom.gov.au/water/about/image/basin-hi_grid.jpg. Accessed 28/12/2022.

Briggs, G. 2021. Magnificent *Mogurnda* from the Atherton Tablelands. Fishes of Sahul 35(2), 1722–1744.

Brown, C., Hammer, M., Unmack, P. & Ebner, B. 2019. *Melanotaenia* sp. nov. 'Malanda'. The IUCN Red List of Threatened Species 2019: e.T123321483A123382516. http://dx.doi.org/10.2305/IUCN.UK.2019-3.RLTS.T123321483A123382516.en

Department of the Environment 2022. *Melanotaenia* sp. nov. ‹Malanda› in Species Profile and Threats Database, Department of the Environment, Canberra. Available from: https://www.environment.gov.au/sprat. Accessed Tue, 15 Nov 2022 11:56:20 +1100.

Department of Environment and Science 2013. Wet Tropics Freshwater Biogeographic Province, WetlandInfo, Department of Environment and Science, Queensland, viewed 13 July 2020, https://wetlandinfo.des.qld.gov.au/wetlands/ecology/aquatic-ecosystems-natural/riverine/wet-tropics/.

Unmack, P.J, Martin, K.C., Hammer, M., Ebner, B., Moy, K. & Brown, C. 2016. Malanda Gold: the tale of a unique rainbowfish from the Atherton Tablelands, now on the verge of extinction. Fishes of Sahul, 30(4), 1039–1054.